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A Gamified Upper Limb Stroke Rehabilitation System with Remote Monitoring and Communication Features

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Abstract - Stroke is one of the main causes of long term disability in adults. Most survivors are left with lasting problems in arm and hand movement that make daily tasks harder to perform. Standard rehabilitation depends on repeated, therapist led exercises, which patients often find hard to continue because they lose interest and cannot always reach a trained clinician outside major hospitals. This paper presents the design and current development status of a low cost, gamified upper limb rehabilitation system that combines a single channel forearm electromyography sensor, connected to a low power microcontroller, with camera based hand tracking. The sensor measures how strongly the forearm muscles are activated, while the camera identifies the position of the hand and wrist during therapeutic movements. A machine learning classifier reads the muscle signal to recognize muscle activity, and the detected hand and wrist gestures are used to control interactive rehabilitation games. A companion web based platform allows patients and physiotherapists to communicate through text and voice messages, and lets physiotherapists remotely monitor patient progress. By combining low cost muscle sensing, engaging games, and remote clinical support, the system aims to improve exercise adherence, provide clear progress data, and bring rehabilitation to patients outside the clinic.

Index Terms - Electromyography, gamification, gesture recognition, hand tracking, mediapipe, remote monitoring, serious games, stroke rehabilitation, telerehabilitation

I. Introduction

Stroke is one of the most common causes of long term disability in adults worldwide [1]. Most survivors are left with lasting problems in arm and hand movement, which make it harder to do activities of daily living (ADLs) such as eating, dressing, and washing.

Good recovery from stroke needs frequent, repeated, and task specific practice of movements, and this is hard to keep up using regular therapy alone [2]. Using sensors and game based technology for rehabilitation has shown promising results, because it can make therapy more engaging, easier to access, and easier to measure [3].

Because the hand and arm are involved in almost every self care and daily living task, from eating and dressing to writing and using a phone, even a partial loss of arm and hand function can affect a stroke survivor's independence more than difficulty walking alone, which can often be helped with a cane or a walker. This makes upper limb recovery one of the most important, and most difficult, goals of stroke rehabilitation.

In recent years, advances in low cost wearable sensors, embedded microcontrollers, webcam based computer vision, and machine learning have opened new ways to support this recovery outside the clinic. Electromyography (EMG) sensors pick up the electrical activity of a muscle even when visible movement is still very small, which makes them well suited to tracking effort in patients whose movement is limited [4]. Webcam based hand tracking, on the other hand, can show the position of the hand and fingers without any wearable device [5]. At the same time, wider use of telehealth, accelerated by the COVID-19 pandemic, has shown that clinicians can meaningfully support patients from a distance when the right tools are available [6].

Bringing these trends together, low cost muscle sensing, camera based hand tracking, and remote clinical support, motivates the system presented in this paper. Before describing the proposed solution, the specific problems with current rehabilitation approaches are described below.

A. Problem description

The current stroke rehabilitation situation has several ongoing problems:

- Low patient motivation: Traditional rehabilitation programs repeat the same exercises again and again, so patients get bored and stop following their exercise plan [2], especially long term stroke patients who need to keep practicing for a long time. Once clinical supervision stops, many patients exercise less and less often.
- Limited access to specialized care: Patients living far from cities often cannot reach rehabilitation centers or trained therapists, which creates a serious gap in their care [7]. Long travel time and cost also stop patients from doing the frequent, short exercise sessions that work best for relearning movement.
- Separate, incomplete technology solutions: Many existing systems only do one thing, for example they track hand gestures or general arm movement, but do not combine game based engagement with clinician monitoring in one easy to use system [8], [9]. Patients and caregivers must then set up several different tools, which makes adoption less likely.
- No clear progress tracking: Regular therapy rarely gives clear, numeric progress data, so it is hard for clinicians to adjust treatment for each patient or track recovery in an organized way. Clinicians must depend on occasional in person checkups or on what the patient remembers to report.
- No combined remote supervision: Tele rehabilitation platforms already exist [7], [6], but few combine a real time game interface, muscle activity sensing, and full clinician monitoring and communication in one system. Patients often have to choose between an engaging home exercise tool and a clinically supervised one.

These problems show the need for one combined, low cost, and easy to set up rehabilitation system. This paper presents such a system, built around a single channel EMG sensor and a standard webcam. The EMG sensor was chosen because it is low cost, easy to wear, and directly measures the muscle activity behind hand and wrist movement. The webcam adds hand and wrist position information without extra wearable sensors, avoiding the setup and calibration that multi sensor motion tracking systems require.

B. Aim and objectives

The aim of this project is to build a low cost rehabilitation game platform that recognizes therapeutic hand and wrist gestures from EMG and webcam data, uses them to control interactive games, and lets physiotherapists monitor and guide patients remotely. The specific objectives are:

1. To develop a hardware subsystem with sensors for the synchronous capture of hand kinematics and muscle activity.
2. To design and implement a feature extraction and classification algorithm to identify hand movements from the collected sensor data.
3. To analyze the accuracy of patient movements using Machine Learning algorithms on the classified movement data, and to develop multiple interactive video games for upper limb rehabilitation of stroke patients, with different gaming exercises.
4. To develop a secure web based platform that enables effective communication between physiotherapists and patients, allows them to monitor patient progress remotely, and provides personalized feedback to patients.

C. Current status of the work

Table 1 summarizes the status of the work, and each part of Sections III and IV begins with a Completed and Future work line. During the build, the camera alone was made to decide game movements, with EMG attached to every command as activity information (Section III-E), and a fixed calibration threshold was replaced by adaptive difficulty (Section III-H).

Table 1. Status of the work

Status	Components
Completed	Battery powered EMG hardware and ESP32 firmware with a USB serial data link; webcam detection of seven movement labels; fusion of camera labels with EMG activity features; four games (Balloon Pop, Car Driving, Treasure Hunt, Shape Matching) with rewards and adaptive difficulty; local application programming interface (API), database and offline session queue; RehabHub physiotherapist web application; EMG dataset from nine healthy volunteers; offline comparison of classical and foundation model EMG classifiers; 47 automated software tests
Future work	Hardware enclosure; Wi-Fi wireless data transmission; patient login, patient–physiotherapist text and voice messaging, and access from outside the clinic computer; use of EMG activity levels in game control; measurement of webcam detection accuracy, latency and usability; more volunteer recordings; data collection from stroke patients after ethics approval

II. Literature Review

A. Wearable sensing for rehabilitation

Wearable sensors have greatly improved how movement is measured in rehabilitation. Inertial measurement units (IMUs) give detailed information about large arm movements [10], [7], while surface EMG sensors record the electrical signal from muscles, which can show a patient's movement intention even when voluntary movement is very limited [11]. A systematic review and meta analysis found that EMG based interventions can improve upper limb motor function after stroke, although evidence that they are better than conventional therapy of

the same amount is still limited [4]. Low cost EMG systems have also been used at home: Tele-REINVENT used a portable EMG interface with games that reward wrist extension, guided by a therapist through video calls, in chronic stroke survivors [6], [12], and MyoGuide evaluated mobile EMG guided wrist extension training in subacute stroke patients [13]. Combining motion sensors with EMG can improve hand and wrist gesture recognition [11], but it adds cost, makes the device harder to wear, and needs extra calibration between sensors. This work therefore uses a single EMG sensor for muscle activity and a webcam, instead of an IMU, for hand position.

B. Serious games for rehabilitation

Serious games improved upper limb function after stroke more than conventional therapy in a meta-analysis of randomized trials [3], robotic games have combined motor and cognitive training [14], and good games follow motor learning rules: task practice, repetition, feedback, and gradually increasing difficulty [15]. Ibrahim et al. combined Kinect tracking, games, and EMG monitoring, but tested only healthy participants [16].

C. Camera tracking, recognition, and foundation models

MediaPipe Hands detects 21 hand landmarks in real time from one RGB camera image [5], and such landmarks have been classified with deep learning to guide home hand exercises [17], but a camera cannot measure how strongly a muscle is activated. Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM) on time domain features are widely used for real time EMG recognition but degrade when the electrode position, arm position, or user changes [18]; random forests are robust to features on different scales [19]; and deep learning needs more data [20]. MOMENT is a family of open transformer models pre-trained on public time series that classify 512 step inputs [21]; they can be used frozen or fine-tuned with low-rank adaptation (LoRA) [22], and they apply reversible instance normalization (RevIN) [23], which removes each window's mean and scale and may remove useful activity level information. Public datasets such as the University of California, Irvine (UCI) EMG Data for Gestures, recorded with an eight channel armband [24], may provide extra training data.

D. Tele rehabilitation and research gap

Tele rehabilitation systems usually combine a sensor, a data link, and a monitoring platform [7], [6], but few combine games, muscle sensing, and therapist communication (Table 2); multi sensor systems [10], [11] are costly to set up at home, and many game systems [14], [16] lack clinician follow-up. It is also unclear how much movement information one low cost EMG envelope carries, and whether foundation models help with such small datasets. This work combines webcam tracking, forearm EMG, adaptive games, and a physiotherapist dashboard, and compares classical classifiers and MOMENT-1 on the same subject splits to decide what role EMG should play.

Table 2. Comparison of related rehabilitation systems

Study	Sensing modality	Serious game	Remote monitoring
Yin and Xu [10]	IMU	Yes	No
Jiang et al. [11]	EMG + IMU	No	No
Heins et al. [14]	Robotic device	Yes	Not reported
Lee et al. [7]	Wearable IMU	No	Yes
Marin-Pardo et al. [6]	EMG	Yes	Yes (video calls)
Ibrahim et al. [16]	Kinect + EMG	Yes	No
Dhuzuki et al. [17]	Camera (MediaPipe)	No	Yes
Proposed system	Webcam (MediaPipe) + EMG	Yes	Partial (clinic dashboard)

III. METHODOLOGY

A. System architecture

Completed: EMG acquisition, webcam detection, fusion, four games, local API and database, and the web application, working together on one laptop. **Future work:** remote access, login, and messaging.

The system has four layers (Fig. 1): sensing (the EMG sensor with an ESP32, and the laptop webcam), processing (the EMG and vision pipelines and the fusion engine), application (four games), and data and monitoring (the local API, a database, and the RehabHub web application). The game process runs an EMG thread, a vision thread, and a fusion thread that every 50 ms combines the latest camera label and EMG features into one command for the game loop (30 frames per second). A FastAPI service with an SQLite database stores patients, sessions, and attempts and serves the web application, and a game that ends while the API is not running saves its session in an offline queue.

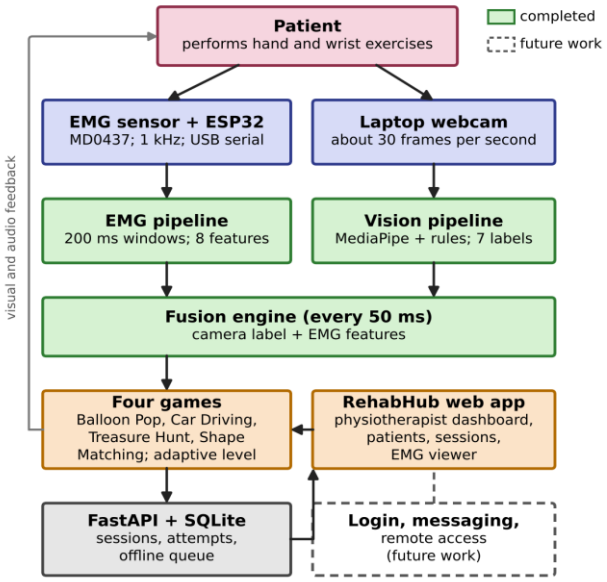


Fig. 1. System architecture (solid: completed; dashed: future work)

B. Hardware implementation

Completed: battery powered EMG hardware, ESP32 firmware, and a USB serial link (Fig. 2, Table 3). **Future work:** an enclosure and wireless transmission over Wi-Fi using the User Datagram Protocol (UDP).

The MD0437 single channel surface EMG module is powered from a split ±9 V battery supply, never from mains. It amplifies, rectifies, and smooths the muscle signal on board, so its output is an activation envelope, with the gain set to stay within the 0 - 3.3 V ESP32 input range. Three Ag/AgCl electrodes are placed as in Table 3. The output goes to general purpose input/output pin 34 (GPIO34) on the first analog-to-digital converter (ADC1) of a 30-pin ESP32, because ADC2 cannot be used with Wi-Fi. The firmware samples the 12-bit ADC at 1 kHz and sends "timestamp (ms), ADC value" lines over USB; invalid lines are discarded, and EMG games pause if no valid sample arrives for 2.5 s.

Table 3. Hardware connections and settings

Item	Connection / setting
Electrodes	Green (+): upper forearm; red (−): 3 - 4 cm towards wrist; yellow (reference): near elbow
EMG module power	+Vs to +9 V, −Vs to −9 V, GND to the 0 V midpoint of the split battery supply
EMG module output	SIG to ESP32 GPIO34 (ADC1); module GND to ESP32 GND
ADC	12-bit (0 - 4095), 11 dB attenuation, 1 kHz sampling in the live firmware
ESP32 to laptop	USB Type-C, serial at 921600 baud; one line per sample: timestamp (ms), ADC value

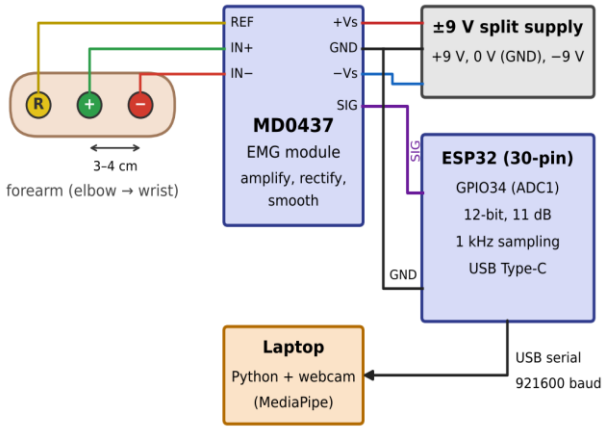


Fig. 2. Wiring of the EMG acquisition hardware

C. Visual hand detection

Completed: real time detection of seven movement labels and a hand pointer. **Future work:** detection accuracy measurement and a learned hand model.

Each webcam frame is processed with the MediaPipe hand landmarker [5] and mapped to seven labels: Rest, Hand Open, Hand Close, Left, Right, Up, and Down (Fig. 3). The steps are:

1. Run the landmarker on the frame and store its capture time; if no hand is found, the label is No Hand.
2. Compute the curl angle of each finger, the fingertip reach divided by the palm size (so that the rules work at any camera distance), the thumb to index pinch ratio, and a full fist flag.
3. Classify the hand shape: a pinch ratio below 0.34, or one or no finger extended with curled fingers and short reach, gives Hand Close (confidence 0.76 - 0.94); three or more extended fingers give Hand Open (0.78 - 0.96); anything else is Rest (0.65). A label is kept only when 3 of the last 5 frames agree.
4. Detect directions from the palm center movement relative to an anchor: more than 2.5% of the frame width gives Left or Right, and more than 4% of its height gives Up or Down; a direction is held for 350 ms with a 280 ms cooldown.
5. Output the label, its confidence, and a smoothed pointer position used by the games for aiming and steering.

D. EMG signal processing and feature extraction

Completed: windowing and eight features, shared by the live system and the offline experiments.

Because the MD0437 output is an activation envelope, the 20 - 450 Hz band pass and 50 Hz notch filters needed for raw EMG are skipped. The signal is divided into 200 ms windows with a 100 ms step, which gives $N = 200$ samples at the 1 kHz live rate and $N = 20$ for the dataset logged at 100 Hz. Eight features are computed for each window: the mean absolute value (MAV), root mean square (RMS), and waveform length (WL) in (1) - (3), the variance, standard deviation, and peak, and the zero crossings and slope sign changes above a noise threshold:

$$\mathrm{MAV} = \frac{1}{N}\sum_{i=1}^{N}|x_i| \tag{1}$$

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}x_i^2} \tag{2}$$

$$\mathrm{WL} = \sum_{i=2}^{N}|x_i - x_{i-1}| \tag{3}$$

where x_i is the i-th envelope sample (ADC value) in the window. The features rise with contraction but are not a calibrated measure of force, so the system never reports them as strength.

E. Fusion of camera and EMG

Completed: fusion engine that attaches EMG activity to every camera command. **Future work:** EMG activity level as an extra condition for game actions.

Every 50 ms, the fusion thread takes the latest camera label, replaces it with Rest when its confidence is below 0.6, takes a majority vote over the last five labels, and attaches the latest EMG window and the frame's capture time (Fig. 3). The earlier design required both signals to agree before a game action, but the results (Section IV-B) show that one forearm EMG channel separates activity levels well and carries little information about direction. EMG therefore never overrides a camera label: it is shown as live activity feedback and stored with every attempt, and the stored capture time allows the camera to game delay to be measured for every command.

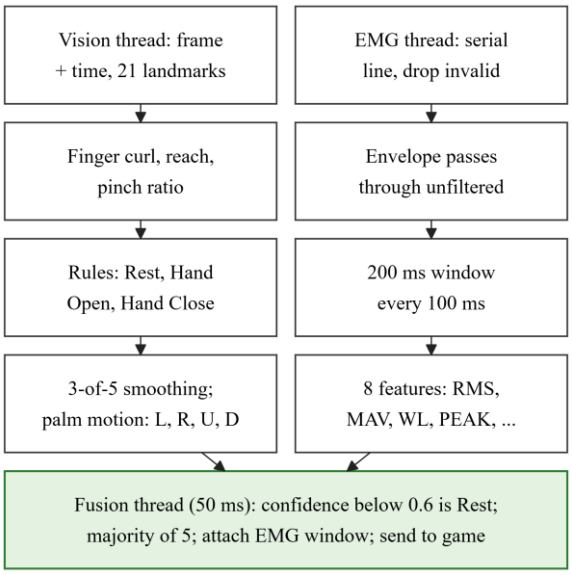


Fig. 3. Vision, EMG and fusion pipelines; each runs in its own thread

F. Dataset

Completed: EMG dataset from nine healthy volunteers with quality checks. **Future work:** re-recording, more volunteers, and stroke patient data after ethics approval.

Nine healthy adult volunteers (S01 - S09), who gave informed consent, each performed the seven movements for about 60 s with the forearm resting on a table, and the labelled envelope was logged at 100 Hz with the 12-bit ADC, giving 61 files, 369,435 samples, 36,873 windows of 200 ms, and 2,494 windows of 5.12 s. Files were checked for missing values, saturation, dropped samples, and joined recordings and split at gaps longer than 1 s; S01 lacks Rest, S04 lacks Down, and S01 lost 26 - 50% of its samples through a lossy link.

G. EMG movement classification

Completed: offline comparison of classical classifiers and MOMENT-1, before and after fine-tuning, with and without public data. **Future work:** per-patient calibration, a second EMG channel, and stroke patient testing.

The classifiers are used only offline, to decide what role EMG should play; none controls the games. Each classifies a window into one of the seven movements on the same subject splits. **Classical models:** LDA, SVM with a radial basis function kernel (C = 10), and a random forest with 200 trees [19] were trained on the eight features, standardized with training subject statistics. **5.12 s windows:** because MOMENT-1 needs 512 evenly spaced samples, each segment was resampled to a 10 ms grid with linear interpolation (S01 was excluded, as 39% of its samples would be interpolated), and each sample x was calibrated per subject with (4), where μ_s and σ_s are the mean and standard deviation of $\log(1 + x)$ over all recordings of subject s (no labels are used):

$$z = \mathrm{clip}\left(\frac{\log(1+x)-\mu_s}{\sigma_s}, -6, 6\right) \quad (4)$$

This keeps the quiet Rest (mean RMS about 38) and the strong Hand Close (about 885) on one scale and removes gain differences between people. Windows of 512 samples were cut every 128 samples, and the eight features were also computed on them.

MOMENT-1: MOMENT-1-small (35.3 million parameters) [21] was used with its untrained head, and as a frozen feature extractor whose embeddings were classified by LDA, SVM, or random forest, with the calibration in (4) or with RevIN [23]; a linear probe trained only the head, and LoRA [22] added rank 8 adapters to the attention projections. **Public data:** UCI windows [24] were converted to envelopes and either added to every training fold (2,120 windows of Rest, Hand Close, and Hand Open) or used to pre-train LoRA. **Evaluation:** because neighbouring windows overlap, data are split by subject. The main test is leave-one-subject-out (LOSO) over S02 - S09, and a holdout split (train S02 - S05, validate S06, test S07 - S09) gives a second view. Chance accuracy is 1/7 = 14.3%. With P_k and R_k the precision and recall of class k and K = 7 classes, the macro F1-score is

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{2P_kR_k}{P_k+R_k} \quad (5)$$

H. Game development

Completed: four Python games with results screens, rewards, and adaptive difficulty (Table 4). **Future work:** EMG activity as a difficulty setting, and patient evaluation.

The games are written with Pygame, launched from the web application, and follow the motor learning principles in Section II-B. Each asks for one target movement at a time and records every attempt with its target, the detected movement, correctness, reaction time, EMG RMS and MAV, and confidence (Table 4). A correct movement scores 10 points, plus 5 during a streak of three, with extra rewards such as a golden balloon (+20). A session lasts 180 s, and the results screen shows score, accuracy, streak, reaction time, and one to three stars. The session accuracy, which is also the movement accuracy (MA) of Objective 3, is

$$\mathrm{MA} = \frac{S}{T} \times 100\% \quad (6)$$

where S is the number of correct attempts and T the total number of attempts. Because the EMG envelope is not a calibrated force (Section IV-B), the planned EMG calibration levels were replaced by difficulty that adapts to accuracy. Levels run from 1.0 to 5.0; after at least four attempts since the last change, the level rises by 0.5 when the accuracy of the last six attempts is 80% or more and falls by 0.5 when it is 40% or less. Each game maps the level linearly to its parameters, for example a new balloon every 1,700 ms at level 1 and every 800 ms at level 5.

Table 4. Games and webcam controls

Game	Webcam control	Trains
Balloon Pop	Move the hand to aim; a full fist on the glowing balloon pops it; open the hand before the next pop	Reaching, hand close and open
Car Driving	Hold the hand left, middle or right to pick one of three lanes and catch stars; a fist brakes	Side to side reach, holding a position, hand close
Treasure Hunt	Push the hand like a joystick to step left, right, up or down on an island; open, then close the hand at each chest	Four directions, hand open and close
Shape Matching	Hover a shape, make a fist to pick it up, carry it, open the hand to drop it in the matching hole	Grasp and release, precise reaching

I. Web application, privacy, and testing

Completed: RehabHub web application, local only operation with coded patient identifiers, and 47 automated tests. **Future work:** login with patient and physiotherapist roles, messaging, encryption, and remote access.

RehabHub, built with React and TypeScript, is used by the physiotherapist. It shows patients, sessions, accuracy over time and by movement, therapy time, and patients needing attention; holds coded patient records with EMG activity by movement and session history; and launches games with a chosen input mode, difficulty, and session length. The API listens only on the local address, no names or camera video are stored, and because there is no login the prototype must not be exposed to a network. The 47 automated tests cover the features, fusion, API, offline queue, difficulty rules, dataset loading, EMG replay, and all four games.

J. Evaluation plan

Future work: all stages below.

First, five to ten healthy volunteers will be used to measure webcam detection accuracy against annotated recordings, game control accuracy using (6), and the camera to game delay (target: median below 300 ms). Second, a usability study with physiotherapists and volunteers will use the System Usability Scale [25] (target: above 68) and interviews. Third, subject to ethics approval, a small pilot with stroke patients will assess tolerance, adherence, and the clinical usefulness of the dashboard.

IV. Results and Discussion

Completed: EMG activity per movement, offline classification results, and functional checks. **Future work:** webcam accuracy, latency, and usability (Section III-J).

A. EMG activity per movement

Table 5 shows the mean window RMS for each movement. Hand Close produces by far the most activity and Rest the least, the four directions lie close together between them, and Hand Open is only about twice Rest. Relative to each volunteer's own Rest, Hand Close ranged from 7.3 times (S03) to 117 times (S02), which shows large differences in gain and electrode contact between people.

Table 5. Mean EMG activity per movement (nine volunteers)

Movement	Mean RMS (ADC units)	Ratio to Rest
Hand Close	884.6	23.0
Up	413.8	10.7
Down	413.5	10.7
Right	387.9	10.1
Left	301.4	7.8
Hand Open	92.0	2.4
Rest	38.5	1.0

B. Classification performance

Table 6 shows all methods on the same splits. With 200 ms windows the best classical model, the SVM, reached 37.2% LOSO accuracy, well above chance but far from usable for game control, and on 5.12 s windows it reached 47.3%. The untrained MOMENT-1 head was at chance, but frozen MOMENT-1 embeddings with a random forest gave the best LOSO result, 48.1% (macro F1 0.443), and 49.3% on the holdout volunteers. Fine-tuning did not beat the frozen features, and replacing the per-subject calibration with RevIN lowered the frozen models by 11 - 13 points.

Table 6. Classification accuracy, seven movements (chance 14.3%)

Method	LOSO acc.	LOSO F1	Holdout acc.
LDA, 200 ms	34.7%	0.297	36.0%
SVM, 200 ms	37.2%	0.321	40.5%
Random forest, 200 ms	34.0%	0.307	37.5%
LDA, 5.12 s	43.8%	0.381	40.3%
SVM, 5.12 s	47.3%	0.425	50.2%
Random forest, 5.12 s	40.3%	0.368	45.0%
MOMENT-1, untrained head	14.2%	0.059	17.4%
Frozen MOMENT-1 + LDA	41.9%	0.392	45.2%
Frozen MOMENT-1 + SVM	46.4%	0.425	48.3%
Frozen MOMENT-1 + random forest	48.1%	0.443	49.3%
Frozen MOMENT-1 + random forest, RevIN	35.3%	0.304	29.2%
MOMENT-1 linear probe	43.0%	0.369	45.5%
MOMENT-1 LoRA	43.2%	0.369	45.5%

C. Per-movement results and public data

Table 7 shows the LOSO recall of each movement. Every method recognized Rest, Hand Open, and Hand Close far better than the four directions; longer windows and MOMENT-1 features mainly improved Hand Open (from 46% to 74%), while Up and Down stayed near chance. Accuracy varied from 16 - 31% for S03 to up to 58.5% for S05 and S06, so one or two point differences

between the best methods are within this spread. Adding the public windows did not raise the best accuracy (frozen MOMENT-1 with a random forest: 47.8%; LoRA: 39.1%), because they cover only three movements, and pre-training LoRA on them gave 44.4%.

Table 7. LOSO recall per movement

Movement	SVM 200 ms	SVM 5.12 s	Frozen MOMENT-1 + random forest	MOMENT-1 LoRA
Rest	81%	74%	78%	72%
Hand Open	46%	59%	74%	74%
Hand Close	66%	79%	75%	76%
Left	23%	47%	40%	32%
Right	9%	40%	31%	18%
Up	19%	23%	9%	19%
Down	11%	5%	27%	3%

D. Implemented system

The authors checked that the sensor, webcam module, fusion engine, four games, database, offline queue, and RehabHub work together, and all 47 automated tests pass. These were functional checks only; the camera to game delay is logged but not yet measured.

E. Discussion and comparison

Using one EMG channel and a webcam instead of IMUs and multi channel EMG keeps the hardware low cost (around LKR 14,000), easy to wear, and free of inertial calibration. A single envelope channel carries reliable information about the activity level, so Rest, Hand Open, and Hand Close are separated well, but little about direction, so the camera decides every game movement and EMG serves as activity feedback and a record of effort. The window length mattered more than the model (the same SVM gained 10.1 points from 200 ms to 5.12 s), a pre-trained foundation model helped without fine-tuning, and per-subject calibration was essential, so a short calibration recording for each patient is a natural next step. Multi channel EMG systems report higher accuracy under controlled conditions [11], but the present 48.1% was obtained with one envelope channel and strict LOSO testing, so a direct comparison is not fair; the results agree with systems that reward the activity of one movement [6] or monitor muscle activity alongside camera tracking [16].

F. Limitations

The classifiers used only nine healthy volunteers, two of whom miss one movement, so the results may not generalize to other users or to stroke patients; a recording labelled as a stroke patient was recently added to the project data, but its origin is still being confirmed, so it is not used. The dataset was logged at 100 Hz while the live firmware samples at 1 kHz, the webcam rules were tuned by hand, latency and usability have not been measured, and the prototype runs on one computer over USB without login or messaging, so remote monitoring is not yet possible.

V. Conclusion

This paper presented a low cost, gamified upper limb stroke rehabilitation system that combines webcam hand tracking, a single channel EMG sensor, four games, and a physiotherapist web application. **Completed:** the battery powered EMG hardware, seven label webcam detection, a fusion engine that attaches EMG activity to every command, four adaptive games, a local API and database, the RehabHub web application, an EMG dataset from nine healthy volunteers, and an offline comparison in which frozen MOMENT-1 features with a random forest reached 48.1% LOSO accuracy over seven movements. **Future work:** login and patient–physiotherapist messaging, Wi-Fi transmission, more volunteers and one sampling rate for training and live use, a calibration recording for each patient and a second EMG channel, and the evaluation in Section III-J, including a pilot with stroke patients after ethics approval. The results show that a single EMG channel is best used for activity feedback while the camera recognizes the movement; they do not yet show clinical benefit.

VI. Ethical Approval

Data were collected from the authors and healthy adult volunteers, who took part voluntarily and gave informed consent. No stroke patient data are used in this paper. Formal ethics committee approval has not yet been obtained, and data collection from stroke patients will start only after approval. The system is a research prototype, not a certified medical device.

VII. Acknowledgements

The authors declare no conflict of interest. AI-assisted tools were used only for language editing and formatting. All scientific content and analysis were developed by the authors.

VIII. References

[1] GBD 2021 Stroke Risk Factor Collaborators, “Global, regional, and national burden of stroke and its risk factors, 1990–2021: A systematic analysis for the Global Burden of Disease Study 2021,” *Lancet Neurol.*, vol. 23, no. 10, pp. 973–1003, 2024, doi: 10.1016/S1474-4422(24)00369-7.

[2] C. Winstein and R. Varghese, “Been there, done that, so what’s next for arm and hand rehabilitation in stroke?” *NeuroRehabilitation*, vol. 43, no. 1, pp. 3–18, 2018, doi: 10.3233/NRE-172412.

[3] I. Doumas et al., “Serious games for upper limb rehabilitation after stroke: A meta-analysis,” *J. NeuroEng. Rehabil.*, vol. 18, no. 1, Art. no. 100, 2021, doi: 10.1186/s12984-021-00889-1.

[4] M. Munoz-Novoa et al., “Upper limb stroke rehabilitation using surface electromyography: A systematic review and meta-analysis,” *Front. Hum. Neurosci.*, vol. 16, Art. no. 897870, 2022, doi: 10.3389/fnhum.2022.897870.

[5] F. Zhang et al., "MediaPipe Hands: On-device real-time hand tracking," arXiv preprint arXiv:2006.10214, 2020.

[6] O. Marin-Pardo et al., "Development of a low-cost, modular muscle–computer interface for at-home telerehabilitation for chronic stroke," *Sensors*, vol. 21, no. 5, Art. no. 1806, 2021, doi: 10.3390/s21051806.

[7] S. I. Lee et al., "Enabling stroke rehabilitation in home and community settings: A wearable sensor-based approach for upper-limb motor training," *IEEE J. Transl. Eng. Health Med.*, vol. 6, pp. 1–11, 2018, doi: 10.1109/JTEHM.2018.2829208.

[8] D. K. Zondervan et al., "Home-based hand rehabilitation after chronic stroke: Randomized, controlled single-blind trial comparing the MusicGlove with a conventional exercise program," *J. Rehabil. Res. Dev.*, vol. 53, no. 4, pp. 457–472, 2016.

[9] X. Song et al., "Cellphone augmented reality game-based rehabilitation for improving motor function and mental state after stroke," in *Proc. IEEE 16th Int. Conf. Wearable Implant. Body Sensor Netw. (BSN)*, 2019, pp. 1–4, doi: 10.1109/BSN.2019.8771093.

[10] Z.-X. Yin and H.-M. Xu, "A wearable rehabilitation game controller using IMU sensor," in *Proc. IEEE Int. Conf. Appl. Syst. Invention (ICASI)*, 2018, pp. 1060–1062, doi: 10.1109/ICASI.2018.8394459.

[11] S. Jiang et al., "Feasibility of wrist-worn, real-time hand, and surface gesture recognition via sEMG and IMU sensing," *IEEE Trans. Ind. Informat.*, vol. 14, no. 8, pp. 3376–3385, 2018, doi: 10.1109/TII.2017.2779814.

[12] O. Marin-Pardo et al., "Functional and neuromuscular changes induced via a low-cost, muscle–computer interface for telerehabilitation: A feasibility study in chronic stroke," *Front. Neuroergon.*, vol. 3, Art. no. 1046695, 2022, doi: 10.3389/fnrgo.2022.1046695.

[13] H.-P. Lin et al., "A usability study on mobile EMG-guided wrist extension training in subacute stroke patients—MyoGuide," *J. NeuroEng. Rehabil.*, vol. 21, no. 1, Art. no. 39, 2024, doi: 10.1186/s12984-024-01334-9.

[14] S. Heins et al., "Robotic-assisted serious game for motor and cognitive post-stroke rehabilitation," in *Proc. IEEE 5th Int. Conf. Serious Games Appl. Health (SeGAH)*, 2017, pp. 1–8.

[15] X. Song et al., "Activities of daily living-based rehabilitation system for arm and hand motor function retraining after stroke," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 621–631, 2022, doi: 10.1109/TNSRE.2022.3156387.

[16] Z. Ibrahim et al., "Enhancing upper limb stroke rehabilitation with electromyography-monitored virtual reality-based games," *J. Robot.*, vol. 2024, Art. no. 8945325, 2024, doi: 10.1155/2024/8945325.

[17] N. H. Mohd Dhuzuki et al., "Design and implementation of a deep learning-based hand gesture recognition system for rehabilitation Internet-of-Things (RIoT) environments using MediaPipe," *IIUM Eng. J.*, vol. 26, no. 1, pp. 353–372, 2025.

[18] Y. Geng et al., "Improving the robustness of real-time myoelectric pattern recognition against arm position changes in transradial amputees," *BioMed Res. Int.*, vol. 2017, pp. 1–10, 2017, doi: 10.1155/2017/5090454.

[19] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[20] M. Panwar et al., "Rehab-Net: Deep learning framework for arm movement classification using wearable sensors for stroke rehabilitation," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 11, pp. 3026–3037, 2019, doi: 10.1109/TBME.2019.2899927.

[21] M. Goswami et al., "MOMENT: A family of open time-series foundation models," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024. [Online]. Available: arXiv:2402.03885

[22] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: arXiv:2106.09685

[23] T. Kim et al., "Reversible instance normalization for accurate time-series forecasting against distribution shift," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[24] S. Lobov et al., "Latent factors limiting the performance of sEMG-interfaces," *Sensors*, vol. 18, no. 4, Art. no. 1122, 2018, doi: 10.3390/s18041122.

[25] J. Brooke, "SUS: A 'quick and dirty' usability scale," in *Usability Evaluation in Industry*, P. W. Jordan et al., Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.